

# Sarek Shakim Mini Project #4 Group: 331-WZ5-55

1) Using Buckingham  $\pi$  theorem rewrite DL relationship between  $\pi$  groups such that:

$$P = f(Q, D, \Omega, \mu, \rho)$$

$\frac{kg}{m^2 s^2}$      $\frac{m}{s}$      $\frac{1}{s}$      $\frac{kg}{m s}$      $\frac{kg}{m^3}$

$$P = f(Q, D, \Omega, \mu, \rho) \Rightarrow [M^1 L^2 T^{-2}] = [L^3 T^{-1}] [L^1] [T^{-1}] [M^1 L^{-1} T^{-1}] [M^1 L^{-3}]$$

a) using  $\rho, \Omega, D$  as P.V.

given  $n = 6 \rightarrow 3$  P.V.  $\Rightarrow 3$   $\pi$  groups

using  $\pi_1, \pi_2, \pi_3 \rightarrow P, Q, \mu$  respectively

$$\begin{aligned} \Rightarrow \pi_1 &= P [D^a] [\Omega^b] [\rho^c] \\ \Rightarrow \pi_2 &= Q [D^a] [\Omega^b] [\rho^c] \\ \Rightarrow \pi_3 &= \mu [D^a] [\Omega^b] [\rho^c] \end{aligned}$$

$$\pi_1 = [M^1 L^2 T^{-2}] = [M^1 L^3 T^{-1}]^a [L^1 T^{-1}]^b [M^1 L^{-3}]^c$$

$$M: 0 = 1 + c \Rightarrow c = -1$$

$$L: 0 = 2 + a - 3c \Rightarrow a = -5$$

$$T: 0 = -2 - b \Rightarrow b = -3$$

$$\Rightarrow \pi_1 = P D^5 \Omega^3 \rho^{-1} = \frac{P}{\rho \Omega^3 D^5}$$

$$\pi_2 = [M^1 L^2 T^{-2}] = [M^1 L^3 T^{-1}]^a [L^1 T^{-1}]^b [M^1 L^{-3}]^c$$

$$M: 0 = c$$

$$L: 0 = 2 + a - 3c \Rightarrow a = -3$$

$$T: 0 = -2 - b \Rightarrow b = -1$$

$$\Rightarrow \pi_2 = Q D^3 \Omega^{-1} \rho^0 = \frac{Q}{\Omega D^3}$$

$$\pi_3 = [M^1 L^2 T^{-2}] = [M^1 L^3 T^{-1}]^a [L^1 T^{-1}]^b [M^1 L^{-3}]^c$$

$$M: 0 = 1 + c \Rightarrow c = -1$$

$$L: 0 = 2 + a - 3c \Rightarrow a = -2$$

$$T: 0 = -2 - b \Rightarrow b = -1$$

$$\Rightarrow \pi_3 = \mu D^2 \Omega^{-1} \rho^{-1} = \frac{\mu}{\rho \Omega D^2}$$

b) using  $Q, \mu, \rho$  as I.V.

given  $n = 6 \rightarrow 3$  P.V.  $\Rightarrow 3$   $\pi$  groups

using  $\pi_1, \pi_2, \pi_3 \rightarrow P, D, \Omega$  respectively

$$\begin{aligned} \Rightarrow \pi_1 &= P [Q^a] [\mu^b] [\rho^c] \\ \Rightarrow \pi_2 &= D [Q^a] [\mu^b] [\rho^c] \\ \Rightarrow \pi_3 &= \Omega [Q^a] [\mu^b] [\rho^c] \end{aligned}$$

$$\pi_1 = [M^1 L^2 T^{-2}] = [M^1 L^3 T^{-1}]^a [M^1 L^{-1} T^{-1}]^b [M^1 L^{-3}]^c$$

$$M: 0 = 1 + a + c \Rightarrow a + c = -1 \Rightarrow a = -1 - c$$

$$L: 0 = 2 + 3a - b - 3c \Rightarrow \boxed{1} \Rightarrow c = 3$$

$$T: 0 = -2 - a - b \Rightarrow a = -3 - b \Rightarrow a = -1$$

$$\Rightarrow \pi_1 = P Q^{-1} \mu^{-4} \rho^3 = \frac{P Q \rho^3}{\mu^4}$$

$$\pi_2 = [M^1 L^2 T^{-2}] = [L^3 T^{-1}]^a [M^1 L^{-1} T^{-1}]^b [M^1 L^{-3}]^c$$

$$M: 0 = b + c \Rightarrow b = -c$$

$$L: 0 = 2 + 3a - b - 3c \Rightarrow 1 + 3a + a - 3c \Rightarrow 1 + a = 0 \Rightarrow a = -1$$

$$T: 0 = -2 - a - b \Rightarrow a = -b = c \Rightarrow a = -1$$

$$\Rightarrow \pi_2 = D Q^{-1} \mu^{-1} \rho^{-1} = \frac{D \mu}{Q \rho}$$

$$\pi_3 = [M^1 L^2 T^{-2}] = [L^3 T^{-1}]^a [M^1 L^{-1} T^{-1}]^b [M^1 L^{-3}]^c$$

$$M: 0 = b + c \Rightarrow b = -c \Rightarrow b = -3$$

$$L: 0 = 2 + 3a - b - 3c \Rightarrow 3(-1) + c - 3c \Rightarrow 3c - 3 + c - 3c \Rightarrow c = 3$$

$$T: 0 = -2 - a - b \Rightarrow a = (-1 - b) \Rightarrow a = (-1 - (-3)) = 2$$

$$\Rightarrow \pi_3 = \Omega Q^2 \mu^{-3} \rho^3 = \frac{\Omega Q^2 \rho^3}{\mu^3}$$

c) Relationship between  $\pi_3$  &  $\pi_1$

$$\pi_1 = \frac{\pi_1 \pi_2}{\pi_3} = \frac{P Q}{\frac{P Q \rho^3}{\mu^4} \frac{D \mu}{Q \rho}} = \frac{P Q \rho^3}{\mu^4} = \frac{P Q \rho^3}{\mu^4}$$

$$\begin{aligned} \boxed{1} \quad 0 &= 2 + 3(-3 - 0 - 6 - 3c) \\ 0 &= 2 - 9 - 18 - 9c \\ &\Rightarrow -25 - 9c \\ &\Rightarrow -25 - 9(-3) \\ &\Rightarrow -25 + 27 = 2 \end{aligned}$$

$$N_2 = \frac{\pi_2}{\pi_2} = \frac{\mu}{\rho \mu D^2} = \frac{\mu \mu D^3}{Q \rho \mu D^2} = \frac{\mu}{Q \rho}$$

$$N_3 = \frac{\pi_3}{\pi_3} = \frac{Q^2}{\mu^2 D^9} = \frac{Q^2 \rho^3 \mu^3 D^5}{\mu^2 \rho^3 \mu^3} = \frac{\rho Q^2 \mu^3}{\mu^3}$$

2) 6 Dimensional Variables:  $\alpha, B, \gamma, \delta, \epsilon, \phi$

$$\begin{aligned} \alpha &= [L^1] \\ B &= [M L^{-3}] \\ \gamma &= [L^2 t^{-1}] \\ \delta &= [L^1] \\ \epsilon &= [M^1 L^2 t^{-3}] \\ \phi &= [t^1] \end{aligned}$$

a) Using  $\delta \in \phi$  as R.V Determine the functional Dependence of  $\alpha$

Given  $n=6$  &  $RV=3 \Rightarrow 3 \pi$  groups

Such that  $\pi_1 = f(\pi_2, \pi_3)$

Using  $\pi_1, \pi_2, \pi_3$  &  $\alpha, B, \gamma$  respectively

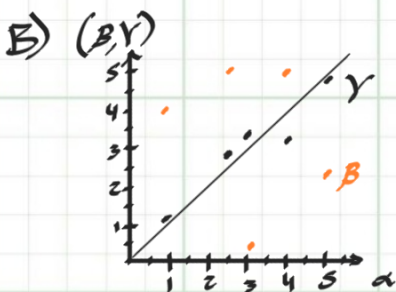
$$\Rightarrow \pi_1 = \alpha [\delta]^a [\epsilon]^b [\phi]^c \quad \pi_2 = B [\delta]^a [\epsilon]^b [\phi]^c \quad \pi_3 = \gamma [\delta]^a [\epsilon]^b [\phi]^c$$

$$\begin{aligned} \pi_1 &= [M^1 L^1 T^0] = [L^1] [L^2 t^{-1}]^a [M^1 L^2 t^{-3}]^b [t^1]^c \\ m: 0 &= b \Rightarrow b=0 \\ L: 1 &= 1+a+2b \Rightarrow a=0 \Rightarrow \pi_1 = \alpha \delta^{-1} = \frac{\alpha}{\delta} \\ T: 0 &= -3b+c \Rightarrow c=0 \end{aligned}$$

$$\begin{aligned} \pi_2 &= [M^1 L^1 T^0] = [M^1] [L^1] [L^2 t^{-1}]^a [t^1]^c \\ m: 0 &= 1+b \Rightarrow b=-1 \\ L: 1 &= -3+a+2b \Rightarrow a=5 \Rightarrow \pi_2 = B \delta^5 \epsilon^{-1} \phi^{-3} = \frac{B \delta^5}{\epsilon \phi^3} \\ T: 0 &= -3b+c \Rightarrow c=-3 \end{aligned}$$

$$\begin{aligned} \pi_3 &= [M^1 L^1 T^0] = [L^2] [L^1] [L^2 t^{-1}]^a [t^1]^c \\ m: 0 &= b \Rightarrow b=0 \\ L: 1 &= 2+a+2b \Rightarrow a=-2 \Rightarrow \pi_3 = \gamma \delta^{-2} \phi^1 = \frac{\gamma \phi}{\delta^2} \\ T: 0 &= -1-3b+c \Rightarrow c=1 \end{aligned}$$

$$\therefore \pi_1 = f(\pi_2, \pi_3) \Rightarrow \frac{\alpha}{\delta} = f\left(\frac{B \delta^5}{\epsilon \phi^3}, \frac{\gamma \phi}{\delta^2}\right) \Rightarrow \alpha = \delta \cdot f\left(\frac{B \delta^5}{\epsilon \phi^3}, \frac{\gamma \phi}{\delta^2}\right)$$



by looking at the scatter plot of the Non dimensional  $B$  &  $\gamma$  vs the Non dimensional  $\alpha$  we can see that the distributions more accurately represent that  $\alpha$  depends on  $\gamma$  and Not on  $B$ .

3) Given:  $\tau_w = f(p, \mu, V, d, \epsilon)$

a)  $\tau_w = N/m^2 = \frac{kg \cdot m/s^2}{m^2} = kg/m \cdot s^2 = [M^1 L^{-1} T^{-2}]$

$p = kg/m^2 = [M^1 L^{-2}]$

$\mu = kg/m \cdot s = [M^1 L^{-1} T^{-1}]$

$V = m/s = [L^1 T^{-1}]$

$d = cm = [L^1]$

$\epsilon = mm = [L^1]$

Using  $p, \mu, V$  as R.V.

$[M^1 L^{-1} T^{-2}] = f([M^1 L^{-2}], [M^1 L^{-1} T^{-1}], [L^1 T^{-1}], [L^1], [L^1])$   $\because d, \epsilon$  both have  $[L^1]$   
 $\because 1 = 5$  w/ 3 R.V.  $\Rightarrow 2 \pi$  groups  $\epsilon$  is a function  $d \Rightarrow \pi_3 = \frac{\epsilon}{d}$  unitless

Using  $\pi_1, \pi_2, \pi_3$  w/  $\tau_w, \mu, \epsilon$  respectively

$\pi_1 = [M^0 L^0 T^0] = [M^1 L^{-1} T^{-2}] [M^1 L^{-2}]^a [L^1]^b [L^1 T^{-1}]^c$

M:  $0 = 1 + a \Rightarrow a = -1$

L:  $0 = -1 - 3a + b + c \Rightarrow -1 + 3 - 2 + b \Rightarrow b = 0$

T:  $0 = -2 - c \Rightarrow c = -2$

$\Rightarrow \pi_1 = \tau_w p^{-1} d^{-2} = \frac{\tau_w}{p d^2}$

$\pi_2 = [M^0 L^0 T^0] = [M^1 L^{-1} T^{-1}] [M^1 L^{-2}]^a [L^1]^b [L^1 T^{-1}]^c$

M:  $0 = 1 + a \Rightarrow a = -1$

L:  $0 = -1 - 3a + b + c \Rightarrow -1 + 3 + b - 1 = 0 \Rightarrow b = -1$

T:  $0 = -1 - c \Rightarrow c = -1$

$\Rightarrow \pi_2 = \mu p^{-1} d^{-1} = \frac{\mu}{p d} \quad * R.E. = \frac{p d V}{\mu}$

$\pi_3 = \frac{\epsilon}{d}$

$\Rightarrow \pi_1 = f(\pi_2, \pi_3) \Rightarrow \tau_w = p d^2 f\left(\frac{\mu}{p d}, \frac{\epsilon}{d}\right)$

b)  $p = 998 \text{ kg/m}^3$   $\mu = 0.001 \text{ kg/m} \cdot s$   $d = 50 \text{ cm} = 0.5 \text{ m}$   $\epsilon = 0.25 \text{ mm} = 0.00025 \text{ m}$

$\Rightarrow \frac{\epsilon}{d} = \frac{0.00025 \text{ m}}{0.5 \text{ m}} = 0.0005 \text{ constant}$

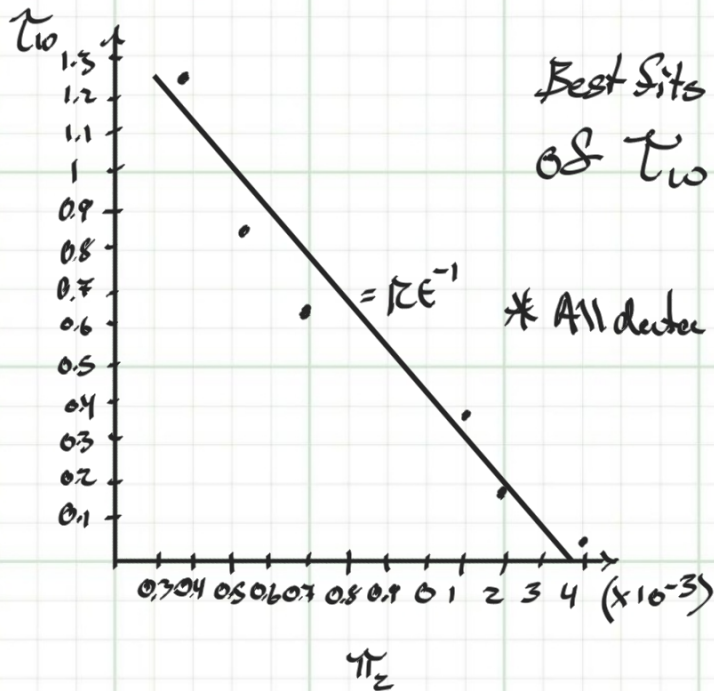
|                             |      |      |      |      |      |      |
|-----------------------------|------|------|------|------|------|------|
| Given: $Q, \text{ gal/min}$ | 1.5  | 3.0  | 6.0  | 9.0  | 12.0 | 14.0 |
| $\tau_w, \text{ Pa}$        | 0.05 | 0.18 | 0.37 | 0.64 | 0.86 | 1.25 |

$\because V = \frac{Q}{A} \Rightarrow \frac{4Q}{\pi D^2} \Rightarrow$  to convert from gal/min  $\rightarrow m^3/s \times 6.769 \times 10^{-5}$

6.769 61964

|                                   |                       |                       |                       |                       |                       |                       |
|-----------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Q, gal/min                        | 1.5                   | 3.0                   | 6.0                   | 9.0                   | 12.0                  | 14.0                  |
| $\tau_w$ Pa                       | 0.05                  | 0.18                  | 0.34                  | 0.64                  | 0.86                  | 1.25                  |
| Velocity $m/s$                    | $4.82 \times 10^{-4}$ | $9.64 \times 10^{-4}$ | $1.93 \times 10^{-3}$ | $2.89 \times 10^{-3}$ | $3.86 \times 10^{-3}$ | $4.50 \times 10^{-3}$ |
| $\pi_1 = \frac{\tau_w}{\rho V^2}$ | 213.7                 | 144.1                 | 99.75                 | 76.68                 | 57.96                 | 61.90                 |
| $\pi_2 = \frac{\mu}{\rho d V}$    | $4.16 \times 10^{-3}$ | $2.08 \times 10^{-3}$ | $1.04 \times 10^{-3}$ | $6.93 \times 10^{-4}$ | $5.70 \times 10^{-4}$ | $4.45 \times 10^{-4}$ |

\*  $\pi_2 = Re^{-1}$



Best fits for entire functional relation  
of  $\tau_w \Rightarrow Re^{-1}$

\* All data points fit this \*

4) Given:  $\rho, \mu, x, g, \sigma, w$

1)  $w = f(\rho, \mu, g, x, d)$

$w = [L^1 t^{-1}]$   
 $\rho = [M^1 L^{-3}]$   
 $\mu = [M^1 L^{-1} T^{-1}]$   
 $g = [L^1 T^{-2}]$   
 $d = [L^1]$   
 $n = 5 + 1 - 3 = 3 = 2 \text{ PI Groups}$   
 using  $\rho, d, g$  R.V

$\pi_1 = [M^0 L^0 T^0] = [L^1 t^{-1}]^a [M^1 L^{-3}]^b [L^1]^{-c} [M^1 T^{-2}]^c$

$M: 0 = a + c \Rightarrow a = -c \Rightarrow a = -1/2$   
 $L: 0 = 1 - 3a - b \Rightarrow 1 - 3(-1/2) - b = 0 \Rightarrow b = 5/2$   
 $T: 0 = -1 - 2c \Rightarrow c = -1/2$   
 $\pi_1 = w \rho^{1/2} d^{5/2} g^{1/2}$

$\pi_2 = [M^0 L^0 T^0] = [M^1 L^{-1} T^{-1}]^a [M^1 L^{-3}]^b [L^1]^{-c} [M^1 T^{-2}]^c$

$M: 0 = a + c \Rightarrow a = -1 - c \Rightarrow a = -1/2$   
 $L: 0 = -1 - 3a + b \Rightarrow b = 1 + 3(-1/2) \Rightarrow b = -1/2$   
 $T: 0 = -1 - 2c \Rightarrow c = -1/2$   
 $\pi_2 = \frac{\mu}{\rho^{1/2} d g}$

$\pi_3 = \frac{x}{d}$

$$\Rightarrow \pi_1 = f(\pi_2, \pi_3) \Rightarrow w = \frac{\sqrt{p}}{\sqrt{\rho g}} \int \left( \frac{\mu}{\sqrt{\rho g}}, \frac{x}{d} \right)$$

assuming steady state

$$ii) \cancel{p} \left( \cancel{\frac{\partial w}{\partial t}} + \cancel{\mu} \frac{\partial w}{\partial x} + \cancel{\nu} \frac{\partial w}{\partial y} + \cancel{\omega} \frac{\partial w}{\partial z} \right) = \cancel{\frac{\partial p}{\partial z}} + \mu \left( \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \rho g_z$$

S.S.      at  $\mu \Rightarrow 0$       only in  $x$

$$\Rightarrow 0 = \mu \frac{d^2 w}{dx^2} + \rho g_z \Rightarrow \rho g_z = \mu \frac{d^2 w}{dx^2} \Rightarrow \frac{d^2 w}{dx^2} = \frac{\rho g_z}{\mu}$$

$$\int \frac{d^2 w}{dx^2} = \frac{\rho g_z}{\mu} \Rightarrow \frac{dw}{dx} = \frac{\rho g_z}{2\mu} x^2 + C_1 x + C_2$$

$$\therefore w(x) = \frac{\rho g_z x^2}{2\mu} + C_1 x + C_2$$

by applying boundary conditions @  $x=0$  +  $x=d$   
Such that  $w(0)=0$   $\frac{dw(x)}{dx} = 0$

$$@ x=0 \quad C_2 = 0$$

$$@ x=d \quad C_1 = -\frac{\rho g_z d}{\mu}$$

$$\text{thus } w(x) = \frac{\rho g_z x^2}{2\mu} - \frac{\rho g_z d}{\mu} x \Rightarrow w(x) = \frac{\rho g_z}{\mu} x \left( \frac{x}{2} - d \right)$$

iii.)

5) Using  $T_P, F_P, F_S, v, d_P$  /  $T_P \propto F_P$

$T_P = K F_P$  \*  $K$  is a proportionality constant

$$T_P = [T]$$

$$F_P = [m^1 L^{-3}]$$

$$v = [m^2 T^{-1}]$$

$$F_S = [m^1 L^{-3}]$$

$$d_P = [L]$$

$$\Rightarrow T_P = K_{F_P} \cdot f(F_S, v, d_P)$$

$$K [T^1 m^1 L^{-3}] = f([m^1 L^{-3}], [m^2 T^{-1}] [L])$$

potential E.V.

$v, d_P \Rightarrow \pi$  groups

6)  $F = f(\rho, D, V, \mu, n)$  Test of 25 cm (model) yield Thrust Data at a wind Vel of 20 m/s  $\mu = \text{constant}$  So test / keep  $\mu$  group out

| Rotation rate (rad/min)                                    | 4800                 | 6000                 | 8000                 |
|------------------------------------------------------------|----------------------|----------------------|----------------------|
| Thrust N                                                   | 6.1                  | 19                   | 47                   |
| Rotation Rate (rad/s)                                      | 80                   | 100                  | 133.3                |
| $\pi_1 = \frac{F \rho}{\rho V^2}$                          | $3.1 \times 10^{-3}$ | $9.7 \times 10^{-3}$ | $2.4 \times 10^{-2}$ |
| $\pi_2 = \frac{\mu d}{\rho V} \Rightarrow \text{Constant}$ |                      |                      |                      |
| $\pi_3 = \frac{n d}{V}$                                    | 1                    | 1.25                 | 1.66                 |

a)

$$\rho = \text{kg/m}^3 = [M^1 L^{-3}]$$

$$\mu = \text{kg/ms} = [M^1 L^{-1} T^{-1}]$$

$$V = \text{m/s} = [L^1 T^{-1}]$$

$$d = \text{cm} = [L^1]$$

$$n = \text{rad/s} = [T^{-1}]$$

$$F = \text{kg} \cdot \text{m/s}^2 = [M^1 L^1 T^{-2}]$$

$\rho, V, D$  as R.V

3  $\pi$  groups

$$\pi_1 = [M^0 L^0 T^0] = [M^1 L^1 T^{-2}] [M^1 L^{-3}]^a [L^1 T^{-1}]^b [L^1]^c$$

$$M: 0 = 1 + a \Rightarrow a = -1$$

$$L: 0 = 1 - 3a + b + c \Rightarrow c = 2 \Rightarrow \pi_1 = F \rho^{-1} V^{-2} d^2 = \frac{F d}{\rho V^2}$$

$$T: 0 = -2 - b \Rightarrow b = -2$$

$$\pi_2 = [M^0 L^0 T^0] = [M^1 L^1 T^{-1}] [M^1 L^{-3}]^a [L^1 T^{-1}]^b [L^1]^c$$

$$M: 0 = 1 + a = a = -1$$

$$L: 0 = -1 - 3a + b + c \Rightarrow c = 1 \Rightarrow \pi_2 = \mu \rho^{-1} V^{-1} d = \frac{\mu d}{\rho V}$$

$$T: 0 = -1 - b \Rightarrow b = -1$$

$$\pi_3 = [M^0 L^0 T^0] = [T^{-1}] [M^1 L^{-3}]^a [L^1 T^{-1}]^b [L^1]^c$$

$$M: 0 = a$$

$$L: 0 = -3a + b + c \Rightarrow c = -b \Rightarrow c = 1 \Rightarrow \pi_3 = n \rho^{-1} V^{-1} d = \frac{n d}{V}$$

$$T: 0 = -1 - b \Rightarrow b = -1$$

a)

